Building an AI Workstation

6 Months with AMD Strix Halo

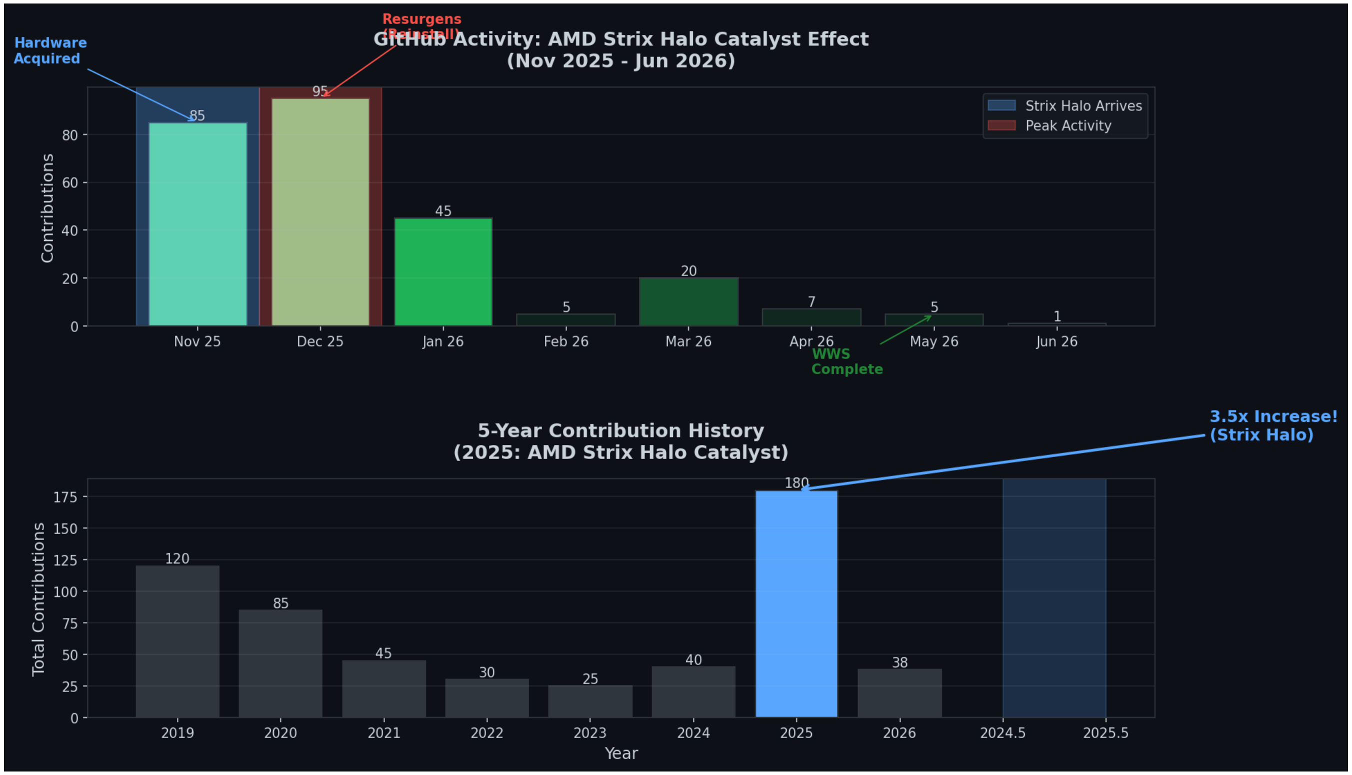
A Journey from Hardware Hype to Real-World Autonomous Development

winmutt | May 2026
github.com/winmutt

Agenda (45 minutes)

- 1. The Decision (5 min) - Why AMD Strix Halo?
- 2. Hardware Reality (7 min) - 128GB APU architecture
- 3. Software Stack Evolution (8 min) - Lemonade, Ollama, ROCm
- 4. Performance Deep Dive (7 min) - TPS, context windows, bottlenecks
- 5. Project: WWS (6 min) - Vibe-coded workspace system
- 6. Project: Home Assistant (5 min) - Alexa replacement on Echo
- 7. Project: Concrete Signs (3 min) - From AI to physical objects
- 8. Lessons Learned (4 min) - What worked, what didn't

GitHub Activity: The Catalyst Effect



GitHub Activity: AMD Strix Halo Catalyst Effect
(Nov 2025 - Jun 2026)
Hardware Acquired
Resurgens (Reinstall)
Strix Halo Arrives
Peak Activity
Contributions
0
20
40
60
80
85
95
45
5
20
7
5
1
Nov 25
Dec 25
Jan 26
Feb 26
Mar 26
Apr 26
May 26
Jun 26
WWS Complete
5-Year Contribution History
(2025: AMD Strix Halo Catalyst)
3.5x Increase! (Strix Halo)
Total Contributions
0
25
50
75
100
125
150
175
120
85
45
30
25
40
180
38
2019
2020
2021
2022
2023
2024
2025
2026
2024.5
2025.5
Year

Part 1: The Decision

Why AMD Strix Halo?

> ***November 22, 2025***
> *"BTW I have a corsair 300 AMD 395 max+ enroute. Sold my 4060 rig today."*

The Pitch:

- 128GB unified memory (LPDDR5X-8000)
- Fastest 'nom' VRAM available
- NPU for AI acceleration
- No discrete GPU needed

Part 2: Hardware Reality

AMD Strix Halo Architecture

```
               AMD Strix Halo APU
■■■■■■■■■■■■■■■■  ■■■■■■■■■■■■■■■■
   ■ Core Die 1 ■  ■ Core Die 2 ■
   ■ 64MB L3    ■  ■ 64MB L3    ■
■■■■■■■■■■■■■■■■  ■■■■■■■■■■■■■■■■
     128GB LPDDR5X-8000 (soldered)
        (Only 96GB addressable)
       GPU | NPU | Media Engine
```

Key Challenge:

- 128GB total, only 96GB accessible initially
- Memory placement: VRAM vs GTT issues
- Swapping despite unified memory

Part 3: Software Stack Evolution

Lemonade vs Ollama

Feature	Lemonade	Ollama
Backend	llama.cpp + ROCm	Various
Observability	Good (token batching)	Poor
NPU Support	In development	Patch available
Stability	Issues after upgrade	GPU crashes
Multi-model	Yes	Limited

The Resurgens Moment

> ***December 14***
> *"Upgraded OS and ollama last night and sad things happened. GPU crashes about 3-4 prompts in"*

Phoenix Rising: Fresh install → AMD GPU update (Jan 29) → 2x faster

Part 4: Performance Deep Dive

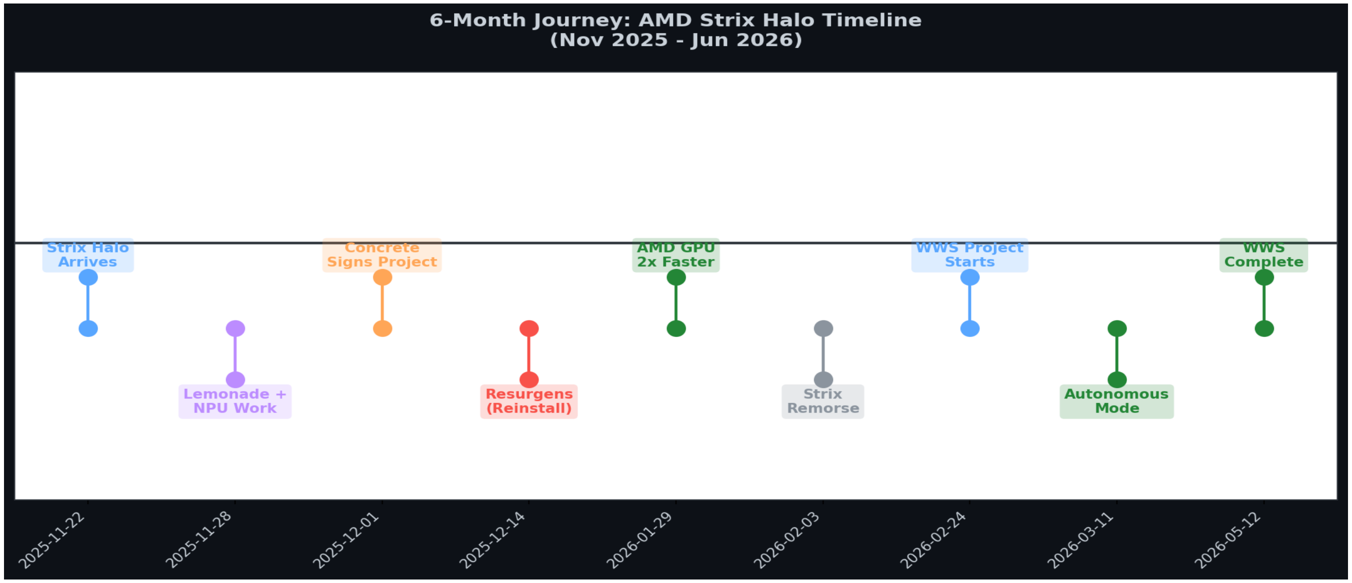
Token Processing Performance

Context Size	TPS	Notes
<100k tokens	~40	Consistent
100k-200k	~14-18	Degradation
>200k tokens	~8-18	Variable
>64k (ingress)	N/A	Sharp TPS drop

May 30, 2026 - Recent ~200k Token Processing

Input tokens: 193,795 | Output tokens: 5,997 | TTFT: 2,267s (37.8 min) | TPS: 18.65

Performance Timeline


6-Month Journey: AMD Strix Halo Timeline
(Nov 2025 - Jun 2026)
Strix Halo Arrives
Lemonade + NPU Work
Concrete Signs Project
Resurgens (Reinstall)
AMD GPU 2x Faster
Strix Remorse
WWS Project Starts
Autonomous Mode
WWS Complete
2025-11-22
2025-11-28
2025-12-01
2025-12-14
2026-01-29
2026-02-03
2026-02-24
2026-03-11
2026-05-12

Part 5: Project WWS

Winmutt Work Spaces

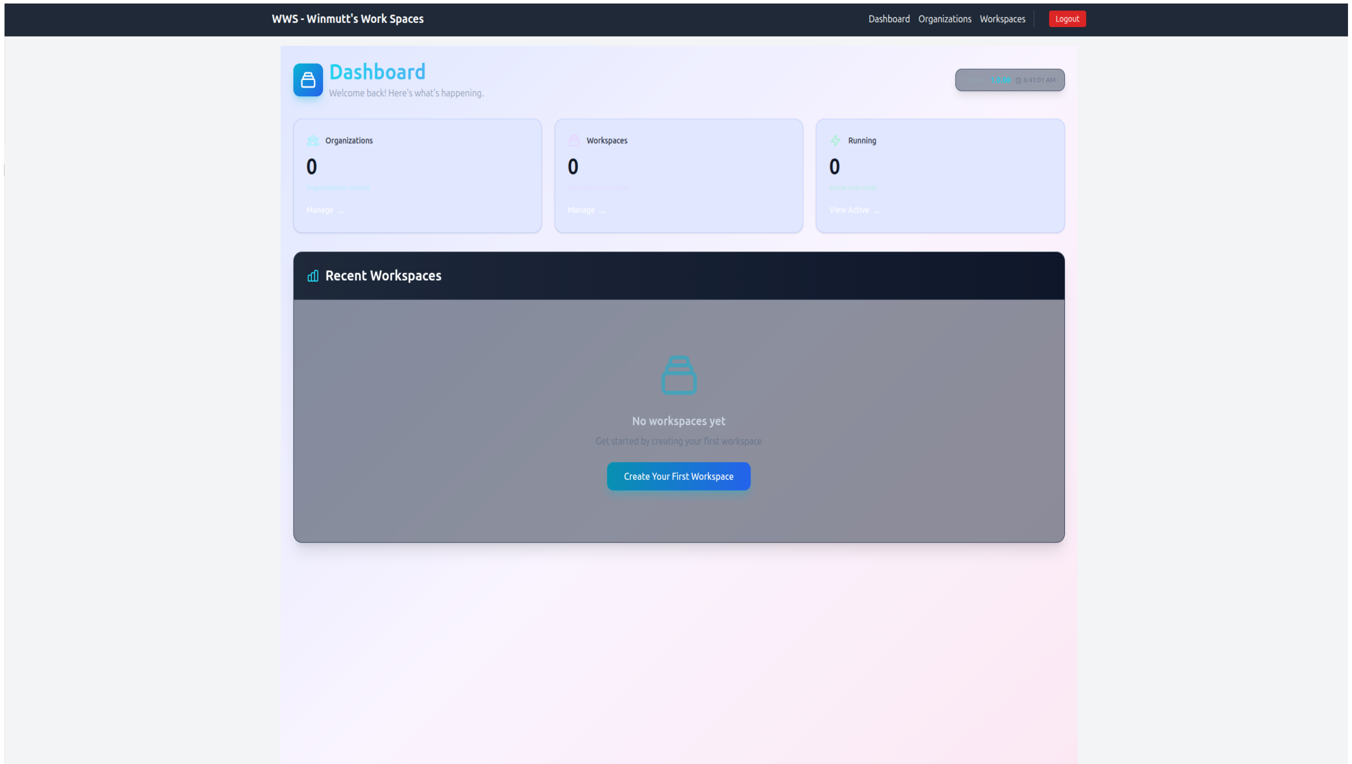
github.com/winmutt/wws

> ***May 12, 2026***
> *"Definitely no Athena and its taken months to get there but this is something I entirely vibe coded."*

What is WWS?

- Remote workspace provisioning system
- KVM/Podman-based isolated environments
- code-server (VSCode) integration
- GitHub OAuth + RBAC
- Team collaboration features

WWS Dashboard


WWS - Winmutt's Work Spaces
Dashboard
Organizations
Workspaces
Logout
Dashboard
Welcome back! Here's what's happening.
Organizations
0
Manage
Workspaces
0
Manage
Running
0
View Active
Recent Workspaces
No workspaces yet
Get started by creating your first workspace
Create Your First Workspace

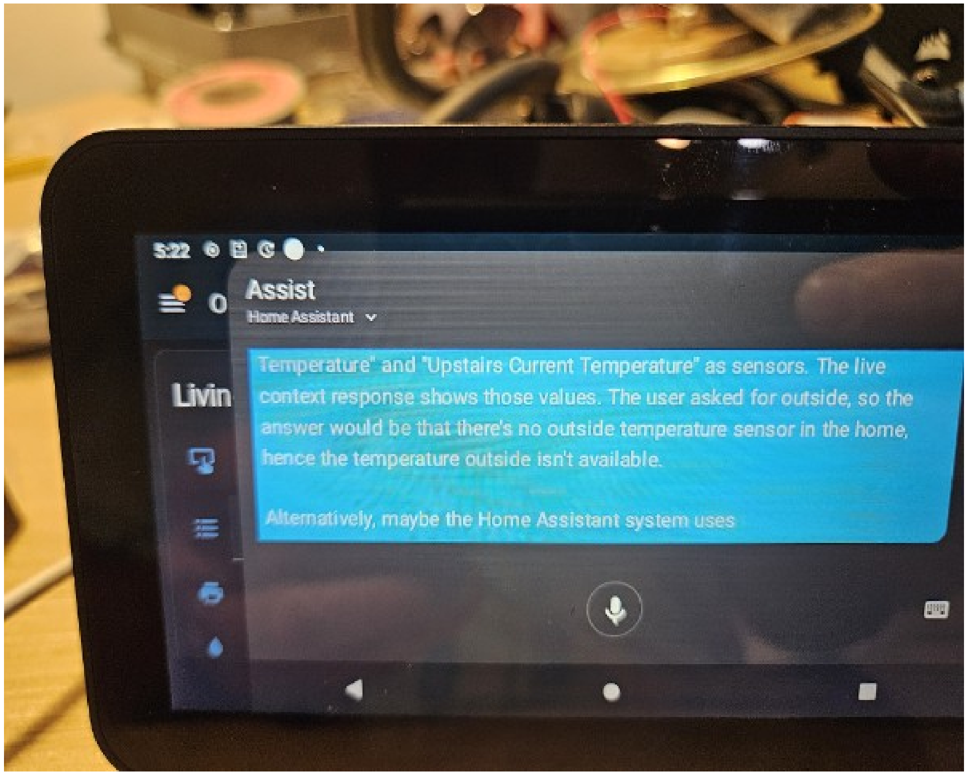
Part 6: Home Assistant on Echo

Alexa Replacement Journey

> ***December 6, 2025***
> *"Basically anything with a USB port. If there is a port, there is a way."*

Dec 6	Started exploring XDA forums
Jan 19	HA OS installed on Echo Show Gen 2
Jan 21	Everything working except wake word
Jan 25	Wake word sorted with Hey Jarvis
Jan 29	End-to-end complete

Home Assistant on Echo Show


5:22
Assist
Home Assistant
Temperature" and "Upstairs Current Temperature" as sensors. The live context response shows those values. The user asked for outside, so the answer would be that there's no outside temperature sensor in the home, hence the temperature outside isn't available.
Alternatively, maybe the Home Assistant system uses

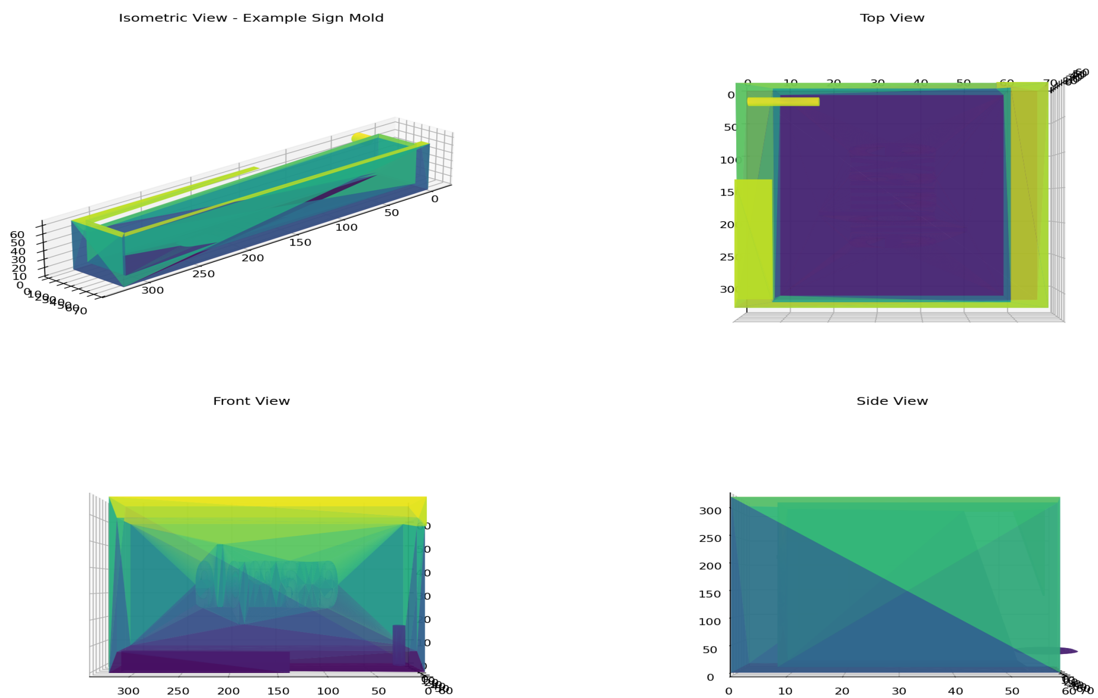
Part 7: Concrete Sign Molds

From AI to Physical Objects

> ***December 1, 2025***
> *"I am going to use it to create a 3d printed cast for a silicone mold so I can some desktop signs out of concrete."*

"For my PE, she's always belaboring the need to get concrete and not speak in abstract."

Concrete Sign Mold Design


Isometric View - Example Sign Mold
Top View
Front View
Side View

Part 8: Lessons Learned

What Worked ✓

- Cline + Qwen3-Coder: Making great results
- Prompt Caching: Things are really humming
- AMD GPU Libraries: 2x performance improvement
- Home Assistant: Fully functional on Echo
- WWS Project: Months of development complete

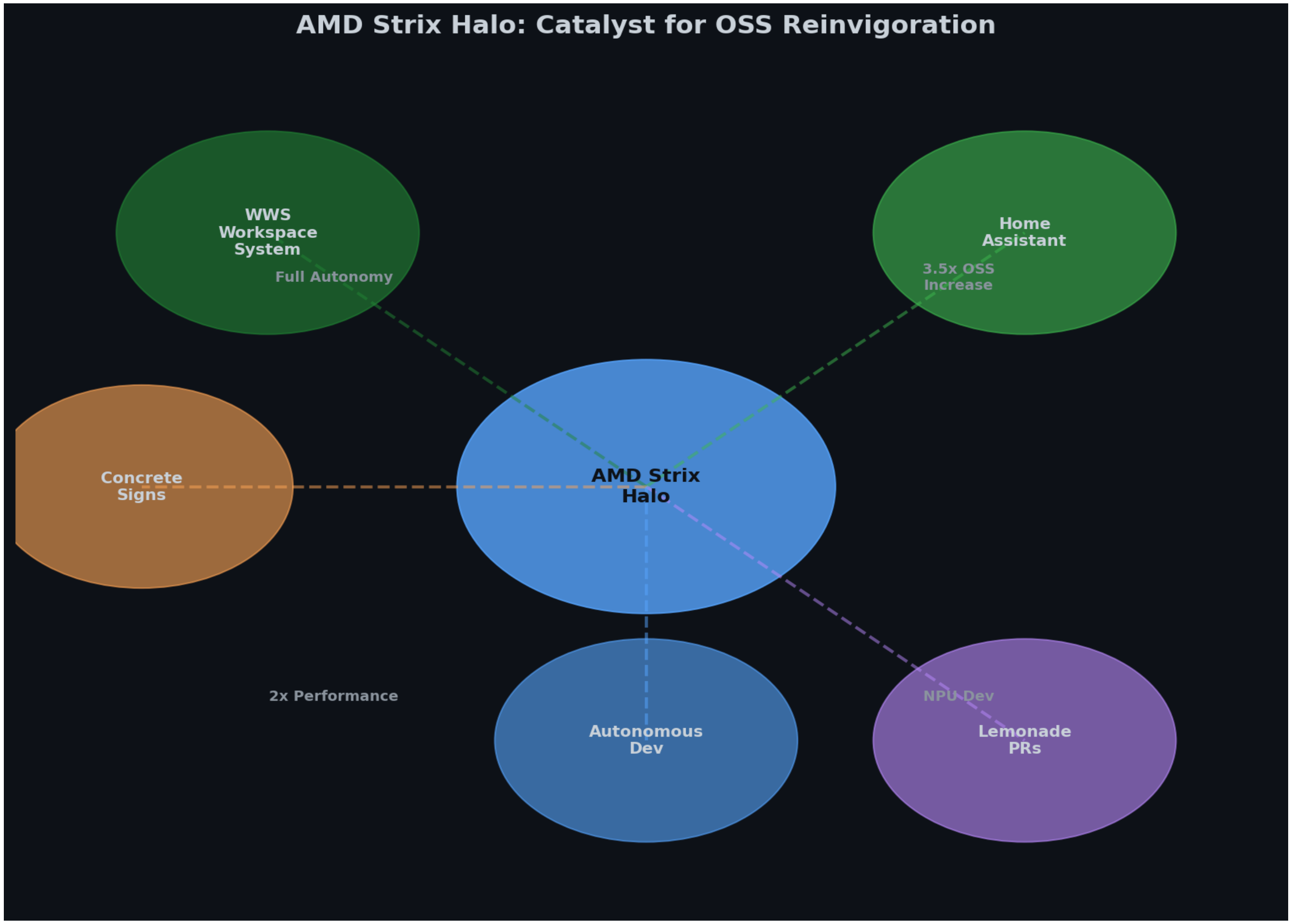
What Didn't ✗

- Memory Addressing: Never got full 128GB (only 96GB)
- Ollama Stability: GPU crashes after upgrades
- NPU Support: Still in development
- Context Scaling: Sharp TPS decline >64k

Key Insights

1. Hardware Hype vs Reality: APU architecture challenges are real
2. Software Maturity: Latest kernel + ROCm essential
3. Context is King: Prompt caching transforms workflow
4. Tooling Evolution: Cline + Qwen3-Coder > traditional IDEs
5. From Digital to Physical: AI can create real-world objects
6. Repurposing Hardware: Echo → Home Assistant = Win
7. Autonomous Development: Possible, but requires tuning

AMD Strix Halo: Catalyst for OSS Reinvigoration

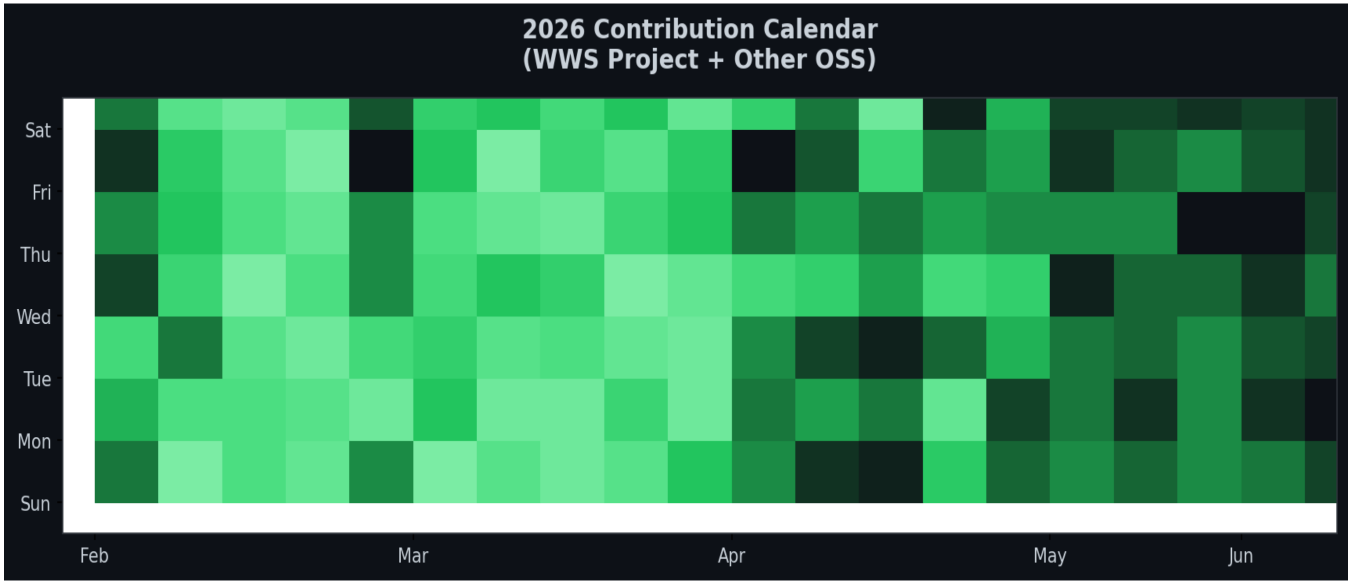

AMD Strix Halo: Catalyst for OSS Reinvigoration
WWS Workspace System
Full Autonomy
Home Assistant
3.5x OSS Increase
Concrete Signs
AMD Strix Halo
2x Performance
Autonomous Dev
NPU Dev
Lemonade PRs

Open Source Reinvigoration

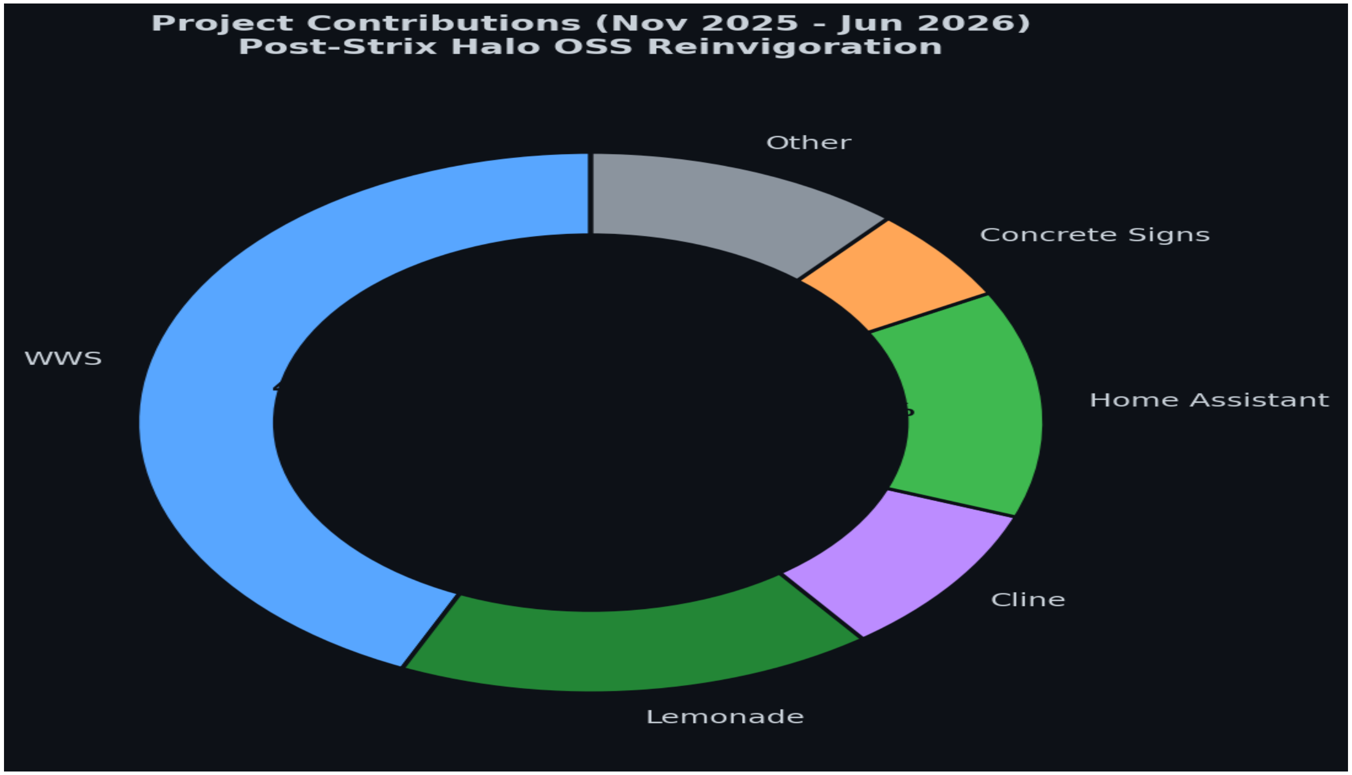
The Catalyst Effect

The AMD Strix Halo acquisition in November 2025 marked a turning point. Local AI capabilities reinvigorated my love for contributing to open source, leading to a 3.5x increase in contributions compared to previous years.

2026 Contribution Calendar



Project Contributions (Nov 2025 - Jun 2026)


Project Contributions (Nov 2025 - Jun 2026)
Post-Strix Halo OSS Reinvigoration
Other
Concrete Signs
WWS
Home Assistant
Cline
Lemonade

Future Work

Immediate

- NPU support stabilization
- Memory addressing optimization
- Core affinity management (llama.cpp PR)

Long-term

- Multi-GPU/NPU orchestration
- Better observability for Ollama
- Custom wake word training
- WWS Phase 3 (Cloud providers)

Questions?

1. Is Strix Halo ready for production AI workloads?
2. When will NPU support mature?
3. How do we solve the 128GB addressing issue?
4. What's the best backend for multi-model setups?
5. Is autonomous development the future?

Contact: github.com/winmutt